

Stereotactic radiotherapy for treatment of juxtapapillary choroidal melanoma: 3-year follow-up

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ABSTRACT

Aims: To report the treatment outcomes and complication rates of stereotactic radiotherapy in the management of patients with juxtapapillary choroidal melanoma.

Methods: A retrospective review of 64 consecutive patients with juxtapapillary choroidal melanoma, located within 2 mm of the optic disc, treated with stereotactic radiotherapy at Princess Margaret Hospital between October 1998 and January 2006.

Results: The median age was 63 years. The median tumour height was 4.2 mm, and median largest basal diameter was 9.8 mm. The prescribed radiation dose was 70 Gy in five fractions over 10 days, and the median follow-up was 37 months. Post-treatment, the actuarial rates of local tumour control, metastases and survival at 37 months were 94%, 15% and 90%, respectively. Actuarial rates of radiation-induced complications at 37 months were neovascular glaucoma 42%, cataract 53%, retinopathy 81% and optic neuropathy 64%. Secondary enucleation was necessary for 10 patients (16%), in four patients for tumour recurrence and in six for painful neovascular glaucoma.

Conclusions: Stereotactic radiotherapy offers a non-invasive alternative to enucleation and brachytherapy in the management of juxtapapillary choroidal melanoma with a high tumour control rate, however, at the expense of a significant rate of long-term ocular complications.

The challenge in dealing with juxtapapillary melanoma lies in the technical difficulty of placing a radioactive plaque securely to cover the entire tumour without producing vision loss through mechanical compromise of the optic nerve, or tilting the plaque away from the optic nerve, which may result in tumour under treatment. In the absence of a clearly accepted management strategy for juxtapapillary choroidal melanomas, various treatments have been described, including: enucleation, notched plaque radiotherapy, charged particle radiotherapy, endoresection and external beam stereotactic radiotherapy (SRT) or radio-surgery.¹⁻⁸ Relatively few reports exist with sufficient follow-up evaluating both efficacy and complications arising from stereotactic radiotherapy in the management of uveal melanoma.¹⁻³ We report our updated experience, with approximately 3-year follow-up data, in treating choroidal melanoma located adjacent to the optic disc with fractionated stereotactic radiotherapy (SRT).

METHODS

A retrospective electronic charts review was conducted of all consecutive patients with juxtapapillary choroidal melanoma, presented between

October 1998 and January 2006 and treated with SRT, after obtaining approval of the hospital's research ethics board.

Patient data included demographics, medical and ocular history, findings of ophthalmological examination, fundus photography and ultrasonography. Systemic evaluation was carried out at the initial visit to rule out any clinical evidence of metastases. Follow-up examinations included ophthalmological exam, fundus photography, ultrasonography and screening for metastases. Visual acuity, tumour measurements and treatment-related complications were documented.

The primary outcome measure was local tumour control. Tumour recurrence was defined as total regrowth of at least 30% of tumour thickness or 500 µm of tumour base, recorded within two successive post-treatment visits. Secondary outcome measures included rates of systemic metastases, survival, secondary enucleation, visual outcome and complications, including radiation cataract, neovascular glaucoma, tumour vasculopathy, retinopathy, vitreous haemorrhage, retinal detachment and optic neuropathy.

Radiation cataract was defined as the development of any lens opacity that was not documented prior to radiation. Neovascular glaucoma (NVG) was diagnosed if neovascularisation of the iris or angle was associated with elevated intraocular pressure. Tumour vasculopathy was defined as the presence of retinal haemorrhage, exudates or vascular occlusion within one disc diameter of the tumour margin. Radiation retinopathy was defined as the presence of retinal haemorrhage, exudates, infarcts, oedema or neovascularisation, more than one disc diameter away from the treated tumour margin, with or without macular involvement. The presence of at least two of the previous findings was required to record the development of radiation retinopathy. Radiation optic neuropathy was defined as optic disc pallor or swelling, with peripapillary haemorrhages not pre-existing.

Visual acuity was presented in lines of vision, where 0 = enucleation, 1 = no light perception, 2 = light perception, 3 = hand movements, 4 = counting fingers. Lines 5 to 16 correspond to Snellen visual acuity ranging from 20/400 to 20/20.

Stereotactic radiotherapy

The total radiation prescribed dose was 70 Gy given in five fractions every alternate day, over 10 days. V70 describes the volume of the globe receiving 70% of the prescribed radiation dose.

All patients were placed in the supine position and immobilised with a Radionics Gill-Thomas-Cosman

stereotactic relocatable head frame (Integra Radionics, Burlington, Massachusetts) adapted with an in-house adjustable eye-monitoring device, consisting of an articulated arm to which a green light-emitting diode (LED) and miniature camera were attached (fig 1). Patients are simulated by computerised axial tomography (fig 2). The patient's non-affected eye stares at the green LED while a camera image is taken to document eye position (reference image). Treatment planning was accomplished using the Radionics XKnife-4, and more recently XKnife RT3, software (Integra Radionics, Burlington, Massachusetts). Multiple converging non-coplanar arcs of radiation were generated using circular collimators ranging from 10 to 27.5 mm. Radiotherapy was delivered using the Radionics circular collimators and linac couch-mount assembly system on a Varian 2100C linear accelerator (Varian, Palo Alto, California), and 6 MV photons. More recently treatment has been delivered using 6 MV photons by the Elekta "Synergy S" linear accelerator (Elekta Oncology Systems, Crawley, UK). During each fraction, the eye position is continuously monitored through the camera image on a television monitor. Alternatively, image-subtraction software compares the eye position during treatment to the reference image, and treatment is interrupted automatically should deviation occur.

Statistical analysis

For continuous factors, statistics were presented as the median and range; for categorical factors, the sample distributions were presented as counts within each category and corresponding proportions. A paired *t* test was applied to compare the difference of tumour height before and after radiotherapy. The patient overall survival was calculated based on Kaplan-Meier estimates. Rates of complications were calculated using the cumulative incidence approach. Risk factors for metastasis were tested using the Cox proportional hazard model. When the variables were continuous, they were dichotomised at the median. A Wilcoxon non-parametric test was applied to compare the initial and final visual acuity with other factors. Results were considered significant if the *p* value was ≤ 0.05 . The statistical analyses were performed using Version 9.1 of the SAS system (SAS Institute, Cary, North Carolina).

RESULTS

Demographic data

Sixty-four patients were included in this study, 36 male (56%) and 28 female (44%), with a median age of 63 years. Medical history included hypertension (28%), cerebrovascular disease

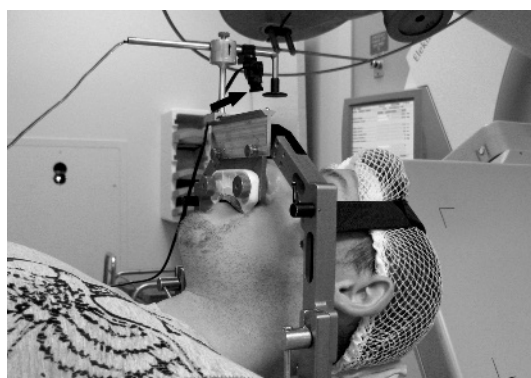


Figure 1 Patient immobilised in stereotactic radiotherapy relocatable head frame with an attached eye monitoring device (arrow).

(6%) and diabetes (5%). Three patients had pre-existing primary open-angle glaucoma. No patients were lost to follow-up.

Tumour characteristics

All tumour margins were located in the juxtapapillary location (0–2 mm). The median distance of the posterior edge of the juxtapapillary tumours and the foveal avascular zone was 0.75 mm (range 0–4.5 mm). The median tumour height was 4.2 mm (range 1.5–11 mm). The median largest base diameter was 9.8 mm (range 4.7–17 mm). Twelve patients (19%) presented with a collar-button appearance. Thirty-one patients (48%) had a localised retinal detachment, and one patient had an extrascleral extension.

Radiation data

The median prescription isodose was 95% (range 90–100%). The median of the maximum doses to the tumour was 74.6 Gy (range 47.2–78.6 Gy), and the median of the minimum doses was 70.2 Gy (range 40.7–74.7 Gy). The median of the maximum doses to the optic nerve was 72.9 Gy (range 60.1–77.2 Gy). The median dose to the lens was 6.7 Gy (1.0–67.5 Gy). The median V70 of the contoured affected eye was 14% (range 1–54%)

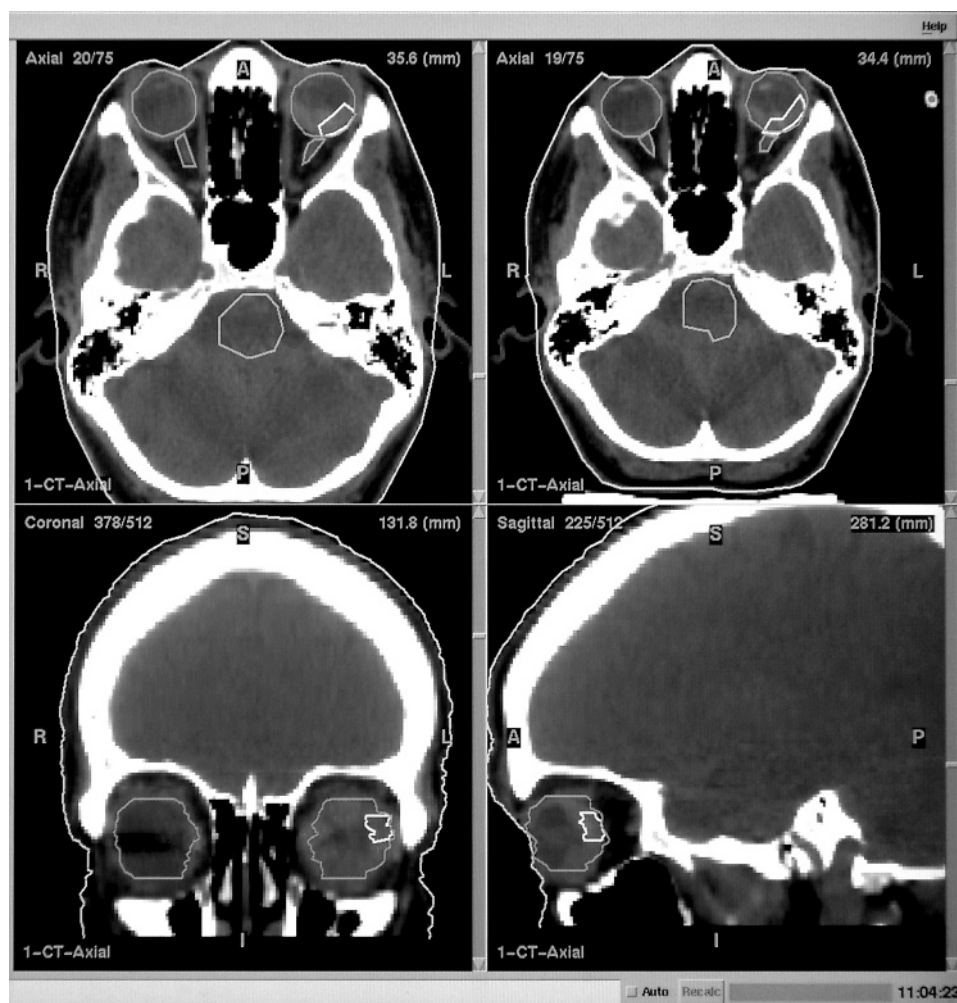
Tumour control

The median follow-up was 37 months (range 6–106 months). Three patients had local tumour recurrence at 6, 7 and 23 months post-SRT. At 37 months, the actuarial local tumour control rate was 94% with 90% CI 87–98% (fig 3). After radiotherapy, the median tumour height decreased by 1.69 mm ($p < 0.0001$). Nine patients developed systemic metastases (seven liver, one lung, one skull base), and the median time to metastases was 20 months (range 5–40 months). The actuarial metastatic rate was 15% (90% CI 8–25%). The initial tumour height was not associated with increased risk of metastasis ($p = 0.11$), while the maximum tumour diameter was significantly associated with increased risk of metastasis ($p = 0.002$). Eight patients died from metastases. The actuarial patient survival rate was 90% (90% CI 80–95%) (fig 4).

Complications

Actuarial rates of all the complications were calculated at a median of 37 months' follow-up. Radiation cataract developed in 53% of patients (90% CI 41–67%) with a median onset of 20 months (range 9–48 months). A trend to a higher rate of cataract development was associated with a greater lens mean dose ($p = 0.07$). The incidence of tumour vasculopathy was 80% (90% CI 71–89%), and a trend to a higher rate of tumour vasculopathy was associated with a higher percentage of the eye receiving 70% of the prescribed radiation dose, V70 ($p = 0.07$). The incidence of radiation retinopathy was 81% (90% CI 72–92%), with a median onset of 16 months (range 2–31 months), and a trend to a higher rate of radiation retinopathy was associated with a greater V70 ($p = 0.13$). The incidence of radiation optic neuropathy was 64% (90% CI 54–78%), with a median onset of 19 months (range 8–40 months). The incidence of neovascular glaucoma (NVG) was 42% (90% CI 32–56%), and the median time to develop NVG was 20 months (range 9–36 months). A higher rate of NVG was associated with greater lens minimum dose ($p = 0.001$). Vitreous haemorrhage occurred in 21 patients, and nine patients had worsening of their retinal detachment. Enucleation was necessary in 10 patients because

Figure 2 Pretreatment localising CT scan.



of tumour recurrence (4/10) and painful neovascular glaucoma (6/10), with a median time to enucleation of 20.5 months.

Visual outcome

Fifteen patients had vision better than 20/40, 22 patients had vision between 20/40 and 20/200, and 27 patients had vision worse than 20/200. Final visual acuity was counting fingers or

less in 51 patients (80%). There was a significant decrease in vision postradiotherapy ($p < 0.0001$).

DISCUSSION

Previous studies have demonstrated no difference in survival by using irradiation versus enucleation in the management of medium-sized uveal melanoma.^{6 9 24} The standard treatment by

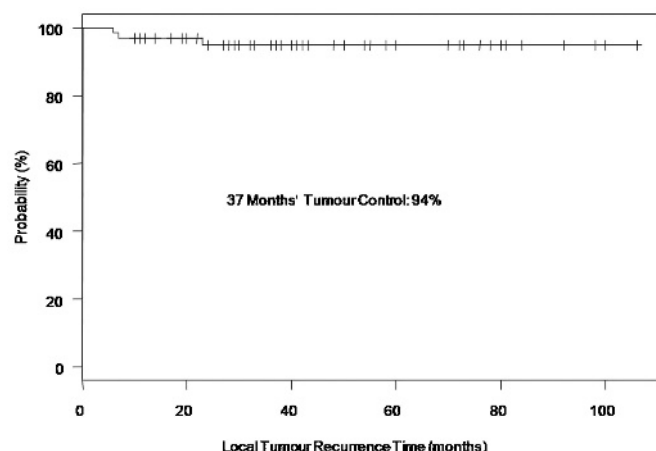


Figure 3 Kaplan–Meier curve showing actuarial tumour local control rate of 94% at 37 months (90% CI 87–98%).

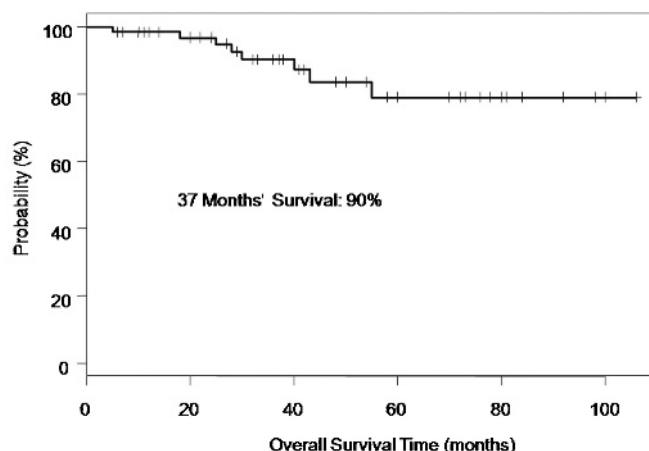


Figure 4 Kaplan–Meier curve showing actuarial survival rate of 90% at 37 months (90% CI 80–95%).

plaque radiotherapy when treating juxtapapillary tumours can be technically difficult, as the optic nerve imposes a physical limitation for accurate insertion of the plaque. This is illustrated in a reported series where the posterior edge of the juxtapapillary plaque was tilted away from the optic nerve, as detected by MRI.¹⁰ Harbour *et al* have reported that by using intraoperative echographic localisation of plaque placement, there was 21% incomplete coverage of at least one margin of the posterior uveal melanomas treated by episcleral plaques, when using the standard localisation techniques.¹¹ This plaque tilt was found to cause reduction greater than 10% in the prescribed radiation dose to the tumour apex in 23% of patients, which may compromise tumour control.²⁰

Notched plaques were developed to accommodate the optic nerve; however, it still carries a potential risk for damage to both the optic nerve and its vasculature during insertion. The use of a “slotted eye plaque” has been recently reported to provide better coverage of juxtapapillary and circumpapillary melanomas.¹²

We elected SRT to treat juxtapapillary melanoma to minimise those risks associated with misplacement of the plaque. This study has shown SRT as an effective management option for uveal melanoma consistent with other reports of radiotherapy (table 1).

SRT is designed to maximise dose to the tumour and minimise dose to surrounding tissues because of the steep dose gradient inherent with radiosurgery. SRT has a practical privilege over particle-based irradiation, such as proton beam and helium ion radiotherapy, in that it spares the patient a surgery for insertion of tantalum rings, required for tumour margin localisation during treatment planning.¹⁵ SRT has advantages over Gamma Knife radiosurgery, since SRT utilises a relocatable immobilisation frame, which permits delivery of irradiation in multiple fractions, while the immobilisation frame employed in Gamma Knife irradiation needs to be pinned to skull bones to achieve complete akinesia during the single-dose radiosurgery. In addition, fractionated radiotherapy theoretically allows for greater repair of the normal tissues within the irradiated field between fractions.

In comparison with SRT, brachytherapy provides more conformal distribution and a lower dose rate of radiation, which theoretically could reduce the late radiation toxicities to nearby structures. Nevertheless, this study demonstrates that the incidence of posterior segment complications such as radiation retinopathy was 81%, and radiation papillopathy was 64%, which approximates the results reported by De Potter *et al*, using ¹²⁵I plaque brachytherapy in management of juxtapapillary choroidal melanoma.⁴ This is possibly the result

of unavoidable lateral scatter of radiation to the nearby retina and optic nerve, irrespective of the method of irradiation to tumours in juxtapapillary location.

With respect to anterior segment complications, our study demonstrated an incidence of NVG of 42%, which is higher than 31% reported incidence with helium ion radiotherapy,^{13 14} 19% with Gamma Knife²⁵ and 15% with proton beam irradiation.^{15 21} It is difficult to extrapolate a reason for this difference; however, current advances in planning techniques with SRT to minimise dose to the anterior segment may reduce the risk of this complication in the future.

The reported incidences of radiation cataract vary among different irradiation studies, perhaps due to the different definitions of radiation cataract among investigators. Generally, a lower incidence of anterior segment complications has been reported with plaque brachytherapy compared with external beam radiotherapy. The placement of the source behind the tumour results in rapid dose-fall-off anteriorly and therefore low dose reaching the anterior segment. In one series of ¹²⁵I plaque radiotherapy for medium-sized melanoma, the incidences of NVG and radiation cataract were 7% and 6%, respectively.¹⁶ In another series using ¹⁰⁶Ru plaques, the incidences of NVG and radiation cataract were 10% and 26% respectively.¹⁷ However, a study of melanomas greater than 8 mm thickness treated by ¹²⁵I plaques reported a higher incidence of NVG (21%) and cataract (66%), which is explained by higher doses exposed to the anterior segment from a closer distance between tumour apex and the anterior ocular structures.¹⁸

The tumour control rate in our series was 94% at 37 months, which is comparable with the reported rates of 85% with brachytherapy treatment of juxtapapillary melanoma^{4 5 19} and 83% with helium ion radiotherapy¹³ at 5 years' follow-up. The tumour control rate in our study, however, was higher than that reported with the conventional Co-60 photon beam irradiation of juxtapapillary medium-sized melanoma,²² which may be due to the more crude localisation techniques used with Co-60 technology. Though our study is retrospective, with less than 5 years' median follow-up, it presents SRT as a non-invasive management option in the treatment of the juxtapapillary uveal melanomas. SRT provides tumour control comparable with other irradiation modalities with a similar median follow-up period. It offers an alternative to enucleation with a high rate of globe retention, but at the expense of long-term higher incidence of ocular complications, particularly to the anterior segment. Ongoing trials for treatment of radiation retinopathy and radiation-induced NVG with retinal photocoagulation, intraocular steroids and early antivascular endothe-

Table 1 Comparison of the treatment outcome and complications reported in various series of radiotherapy of uveal melanoma

	De Potter <i>et al</i> ⁴	Summanen <i>et al</i> ¹⁷	Char <i>et al</i> ¹³	Roberge <i>et al</i> ²²	Lumbroso <i>et al</i> ²¹	Moderati <i>et al</i> ²³	This study
Radiation type	Iodine-125 (85 Gy)	Ruthenium-106 (100 Gy)	Helium ion (50–80 Gy)	External Co-60 (70 Gy)	Proton beam (60 Gy)	Gamma knife (30–50 Gy)	Stereotactic (70 Gy)
Melanoma location	Juxtapapillary	Not specified	Not specified	Juxtapapillary	Not specified	Not specified	Juxtapapillary
Follow-up	78 ms	31 ms	61 ms	60 ms	38 ms	31 ms	37 ms
No of patients	93	100	164	26	1062	78	64
Tumour control	85%	94%	93%	84%	97%	91%	94%
Secondary enucleation	22%	10%	16%	19%	8%	10%	16%
Metastases	12%	—	18%	26%	15%	10%	15%
Radiation cataract	—	26%	41%	8%	22%	6%	53%
Neovascular glaucoma	—	10%	31%	27%	15%	19%	42%
Radiation retinopathy	87%	24%	46%	46%	47%	47%	81%
Radiation papillopathy	52%	10%	—	—	15%	16%	64%

lial growth factor (anti-VEGF) injections²⁵ are promising to potentially reduce some of those radiation induced complications.

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REFERENCES

1. Emara K, Weisbrod DJ, Shahgal A, *et al.* Stereotactic radiotherapy in the treatment of juxtapapillary choroidal melanoma: preliminary results. *Int J Radiat Oncol Biol Phys* 2004;**59**:94–100.
2. Bellman C, Fuss M, Holz FG, *et al.* Stereotactic radiation therapy for malignant choroidal tumors. *Ophthalmology* 2000;**107**:358–65.
3. Dieckmann K, Georg D, Bogner J, *et al.* Optimizing Linac-based stereotactic radiotherapy of uveal melanomas: 7 years' clinical experience. *Int J Radiat Oncol Biol Phys* 2006;**66**:47–52.
4. De Potter P, Shields CL, Shields JA, *et al.* Plaque radiotherapy for juxtapapillary choroidal melanoma: Visual acuity and survival outcome. *Arch Ophthalmol* 1996;**114**:1357–65.
5. Lommatzsch PK, Alberti W, Lommatzsch R, *et al.* Radiation effects on the optic nerve observed after brachytherapy of choroidal melanomas with ¹⁰⁶Ru/¹⁰⁶Rh plaques. *Graefes Arch Clin Exp Ophthalmol* 1994;**232**:482–7.
6. De Potter P, Shields CL, Shields JA, *et al.* Impact of enucleation versus plaque radiotherapy in the management of juxtapapillary choroidal melanoma on patient survival. *Br J Ophthalmol* 1994;**78**:109–14.
7. Lommatzsch PK, Lommatzsch R. Treatment of juxtapapillary melanomas. *Br J Ophthalmol* 1991;**75**:715–17.
8. Seddon JM, Gragoudas ES, Egan KM, *et al.* Uveal melanomas near the optic disc or fovea: Visual results after proton beam irradiation. *Ophthalmology* 1987;**94**:354–62.
9. Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: V Twelve-year mortality rates and prognostic factors: COMS report No 28. *Arch Ophthalmol* 2006;**124**:1684–93.
10. Williams D, Mieler W, Jaffe G, *et al.* Magnetic resonance imaging of juxtapapillary plaques in cadaver eyes. *Br J Ophthalmol* 1990;**74**:43–6.
11. Harbour JW, Murray TG, Byrne SF, *et al.* Intraoperative echographic localization of iodine 125 episcleral radioactive plaques for posterior uveal melanoma. *Retina* 1996;**16**:129–34.
12. Finger PT. Finger's "Slotted" Eye Plaque for radiation therapy: treatment of juxtapapillary and circumpapillary intraocular tumors. *Br J Ophthalmol* 2007;**91**:891–4.
13. Char DH, Castro JR, Kroll SM, *et al.* Five-year follow-up of helium ion therapy for uveal melanoma. *Arch Ophthalmol* 1990;**108**:209–14.
14. Char DH, Kroll SM, Castro J. Ten-year follow-up of helium ion therapy for uveal melanoma. *Am J Ophthalmol* 1998;**125**:81–9.
15. Gragoudas ES, Seddon JM, Egan K, *et al.* Long-term results of proton beam irradiated uveal melanomas. *Ophthalmology* 1987;**94**:348–53.
16. Shields CL, Cater J, Shields JA, *et al.* Combined plaque radiotherapy and transpupillary thermotherapy for choroidal melanoma: tumor control and treatment complications in 270 consecutive patients. *Arch Ophthalmol* 2002;**120**:933–40.
17. Summanen P, Immonen I, Kivela T, *et al.* Radiation related complications after ruthenium plaque radiotherapy of uveal melanoma. *Br J Ophthalmol* 1996;**80**:732–9.
18. Shields CL, Naseripour M, Cater J, *et al.* Plaque radiotherapy for large posterior uveal melanomas (> or = 8-mm thick) in 354 consecutive patients. *Ophthalmology* 2002;**109**:1838–49.
19. Karlsson UL, Augsburger JJ, Shields JA, *et al.* Recurrence of posterior uveal melanoma after ⁶⁰Co episcleral plaque therapy. *Ophthalmology* 1998;**96**:382–8.
20. Almony A, Breit S, Zhao H, *et al.* Tilting of radioactive plaques after initial accurate placement for treatment of uveal melanoma. *Arch Ophthalmol* 2008;**126**:65–70.
21. Lumbruso L, Levy C, Plancher C, *et al.* Results of proton beam irradiation for treatment of choroidal melanoma. *J Fr Ophthalmol* 2002;**25**:290–7.
22. Roberge D, Nordstorm S, Corriveau C, *et al.* Treatment of medium-sized juxtapapillary melanoma with external Co-60 photon therapy. *Radiother Oncol* 2005;**74**:71–3.
23. Modorati G, Miserocchi E, Galli L, *et al.* Gamma knife radiosurgery for uveal melanoma: 12 years of experience. *Br J Ophthalmol* 2009;**93**:40–4.
24. Seddon JM, Gragoudas ES, Albert DM, *et al.* Comparison of survival rates for patients with uveal melanoma after treatment with proton beam irradiation or enucleation. *Am J Ophthalmol* 1985;**99**:282–90.
25. Finger PT, Chin K. Anti-vascular endothelial growth factor bevacizumab (avastin) for radiation retinopathy. *Arch Ophthalmol* 2007;**125**:751–6.